

PROVISIONAL PATENT APPLICATION

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

TITLE: DATABASE-ENFORCED HUMAN GOVERNANCE SYSTEM FOR AI-GENERATED ARTIFACTS **APPLICANT:** Identity Nails LLC **INVENTOR:** Christina Cabral

1. FIELD OF THE INVENTION

The invention relates to systems for governing artificial-intelligence (AI) outputs by **architecturally enforcing** governance via automated gates implemented at the database persistence layer.

2. BACKGROUND OF THE INVENTION

Existing AI governance systems fail to prevent unauthorized state transitions because enforcement mechanisms operate outside the data persistence layer, allowing artifacts to reach executable states through alternative code paths or metadata manipulation. No prior system architecturally constrains state transitions at the database engine level. The present invention solves this technical problem by implementing automated enforcement gates at the database architecture itself, rendering circumvention **architecturally constrained to prevent unauthorized state transitions**.

3. BRIEF DESCRIPTION OF THE DRAWINGS

- **FIG. 1** illustrates a database-enforced sovereignty gate.
- **FIG. 2** illustrates an append-only reasoning ledger with intent anchoring.
- **FIG. 3** illustrates JSON canonicalization and signature validation.
- **FIG. 4** illustrates forensic purge certification with hash-before-delete.
- **FIG. 5** illustrates composite governance with multiple artifacts.
- **FIG. 6** illustrates violation watermark injection and downstream detection.

4. DETAILED DESCRIPTION OF THE DRAWINGS

- **FIG. 1: Sovereignty Gate (100):** An Artifact State (104) transition is evaluated by a persistence-layer Boolean gate (102) that verifies approval conditions (106), blocking unauthorized **External Service (108)** requests until producing a **Finalized Artifact (110)**.

- FIG. 2: Reasoning Ledger (200): Illustrates intent anchoring where an Intent Frame (206), sampled at a predetermined sampling interval, generates a Confidence Score (204) associated with a **Reasoning Entry (202)** in an append-only structure culminating in **Human Approval (208)**.
 - FIG. 3: Payload Security (300): Details Canonicalization (302) and Signature Verification (304) to prevent unauthorized state manipulation.
 - FIG. 4: Forensic Purge (400): Depicts recording a cryptographic fingerprint (404) in an Immutable Audit Log (408) prior to the Deletion Event (406).
 - FIG. 5: Composite Governance (500): Subjecting distributed AI actions to structural gate enforcement (506).
 - **FIG. 6: Violation Watermark (600)**: Depicts a defense-in-depth mechanism where the Sovereignty Gate (102) triggers a Watermark Injection Module (602) upon detecting unauthorized transitions. A Violation Signature (604) is computed and injected into artifact metadata (606), creating a persistent Violation Watermark (608) detectable by downstream systems (610).
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5. EXEMPLARY CLAIMS (TRINITY FORTRESS + SCARLET LETTER)

1. Architectural Enforcement Claim: A system for architecturally enforcing governance over AI-generated artifacts, comprising:

- An **automated enforcement gate** implemented at a database persistence layer;
- Wherein said gate physically blocks an AI artifact from transitioning to a finalized state unless a database constraint evaluates to **TRUE** by cryptographically verifying that a human approval record—bearing a timestamped digital signature binding the approval to a specific reasoning sequence—exists and is immutably associated with said artifact at the persistence layer.

2. Intent Reconstruction Claim: The system of claim 1, further comprising an intent reconstruction module configured to:

- Sample operator interaction data at a **predetermined sampling interval** sufficient to capture human creative emphasis;
- Generate a **confidence score** by comparing said interaction data to AI reasoning steps via a **perceptual similarity metric** that produces a normalized confidence score;
- Wherein the enforcement gate of claim 1 rejects the artifact transition if the confidence score fails to meet a **governance threshold**.

2A. The system of claim 2, wherein the perceptual similarity metric comprises **perceptual hashing** that generates discrete hash values from visual frames, and wherein similarity is computed by Hamming distance between hashes.

3. Forensic Compliance Claim: The system of claim 1, further comprising a forensic non-possession protocol that:

- Computes a cryptographic fingerprint of source material prior to a deletion event;
- Records said fingerprint in a **tamper-evident audit log** structurally bound to the governed artifact to provide persistent evidence of destruction without retention of sensitive source data.

4. Ghost-Path Defense Claim: The system of claim 1, wherein the enforcement gate is configured as a **database trigger firing on all state-bearing operations**—including inserts, updates, partial metadata modifications, **server-side functions, materialized views, and replicated tables**—to prevent unauthorized state propagation through secondary data paths.

5. Violation Watermark Claim: The system of claim 1, wherein the enforcement gate is further configured to execute a **watermark injection protocol** comprising:

- (a) Detecting an unauthorized state transition attempt;
- (b) Computing a **violation signature** comprising a cryptographic hash binding the artifact identifier, transition timestamp, and breach classification;
- (c) Injecting said signature into a metadata field of the artifact prior to blocking the transaction;
- (d) Wherein said signature persists even if the transaction is subsequently forced through privilege escalation or database-level override;
- (e) Such that any downstream system can detect the signature and implement secondary enforcement (execution refusal or forensic logging).